MIL-D-50727B(MU)
11 June 1973
SUPERSEDING
MIL-D-50727A(MU)
21 October 1971

MILITARY SPECIFICATION

DELAY ELEMENT ASSEMBLIES

1. SCOPE

1.1 This specification covers the production, assembly and inspection requirements for Delay Element Assemblies. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-9	- Screw Threads Conventions and Method of Specifying
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-171	- Finishing of Metal and Wood Surfaces

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Applicable Drawings	- Delay Element Assembly (See 6.1 and 6.3)
C9266885	- Container, Metal Packing Ammunition for Delay Element Assembly 0.10 Second Delay, 0.30 Second Delay, 0.60 Second Delay, 0.85 Second Delay, or 1.0 Second Delay
C9266886	- Box, Packing Ammunition for Delay Element Assembly 0.10 Second Delay, 0.30 Second Delay, 0.60 Second Delay, 0.85 Second Delay or 1.0 Second Delay

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N. Y. 10018).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the drawings pertaining to the applicable Delay Assembly specified in the invitation for Bid, Request for Proposal, contract or other procurement documents (see 6.1 and 6.3) and referenced specifications.

3.2 Protective finish.-When protective finish is required by component details, finish requirements shall be as required on the drawings and reference specifications MIL-C-60539 and MIL-STD-171 and shall apply to all surfaces regardless of contour configurations.

3.3 Vibration.-The delay element assembly shall withstand vibrations as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer or delay charge or derangement of components.

3.4 Primer sensitivity.-The primer when crimped and sealed in a retainer assembly shall fire when drop tested at an energy level of 26.0 inch - ounces and shall not fire when drop tested at an energy level of 4.0 inch - ounces.

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3.5 Functional.

3.5.1 The delay element shall not fail to fire.

3.5.2 Ballistics.-The delay charge shall produce the delay time required by the applicable assembly drawing.

3.6 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defect.-The metal parts shall be free from cracks, splits, inclusions, porosity or other metal defects.

3.6.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First Article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 100 primer retainer assemblies for primer sensitivity test, 30 delay element assemblies for vibration and ballistic tests and 10 sets of unassembled components for visual and dimensional examination and tests.

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4.2.1 (Cont'd)

The sample shall be manufactured, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and Tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

TABLE I

Temperature	Vibrated	Not Vibrated
-65° to -70°F	4	6
65° to 75°F	4	6
200° to 205°F	4	6

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-An ammunition lot shall consist of an identifiable group of Delay Element Assemblies consisting of:

a. Components produced by a homogeneous process.

b. Primers from not more than one lot.

c. Delay charge from not more than one lot.

d. Output charge from not more than one lot.

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4.3.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ANSI-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

	Total Allowable Defects per Sample Size			
Sample Sizes for Major Defects	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Sizes for Minor Defects	110	150	225	300
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.3.2.1 Body, Drawing D11731623, covering a detail of the applicable Delay Element Assembly Drawing.

	DEFECT	METHOD OF INSPECTION
CRITICAL		
	Metal defective	Visual
MAJOR		
101.	Pitch diameter of internal threads, max.	Gage
102.	Pitch diameter of external threads, min.	Gage
103.	Minor diameter of internal threads, max.	Gage
104.	Major diameter of external threads, min.	Gage
105.	Concentricity of pitch diameter of external thread with counter-bore	Gage
106.	True position of long cavity with counter-bore	Gage
107.	Concentricity of flash hole with counter-bore	Gage
108.	True position of small diameter with counterbore	Gage
109.	True position of large diameter with counterbore	Gage
110.	True position of small diameter of tapered bore with pitch diameter of internal thread	Gage
111.	True position of large diameter of tapered bore with pitch diameter of internal thread	Gage
112.	Concentricity of pitch diameter of internal thread with flash hole	Gage
113.	Diameter of counterbore, max.	Gage
114.	Diameter of long cavity, max.	Gage
115.	Large diameter of tapered bore, max.	Gage
116.	Depth of counterbore, max.	Gage
117.	Depth of long cavity, min.	Gage
118.	Depth of angle counterbore from face of body	Gage
119.	Length from face of small outside diameter to shoulder	Gage
120.	Symmetry of wrench flats with large outside diameter	Gage
121.	Angle at bottom of tapered bore	Gage

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4.3.2.1 (Cont'd)

	Defect	Method of Inspection
122.	Length of internal thread, from face max.	Gage
123.	Length of internal thread from face, min.	Gage
124.	Diameter of flash hole	Gage
125.	Protective finish inadequate or defective	Visual
126.	Burr	Visual
127.	Foreign matter	Visual
128.	Length from face to bottom of tapered bore	Gage
129.	Surface finish on tapered bore	Visual

MINOR

201.	Pitch diameter of internal thread, min.	Gage
202.	Pitch diameter of external thread, max.	Gage
203.	Diameter of counterbore, min.	Gage
204.	Diameter of long cavity, min.	Gage
205.	Diameter of small end	Gage
206.	Diameter of body	Gage
207.	Large diameter of tapered bore, min.	Gage
208.	Depth of counterbore, min.	Gage
209.	Depth of long cavity, max.	Gage
210.	Dimension over flats	Gage
211.	Width of undercut	Gage
212.	Diameter of undercut, min.	Gage
213.	Length from face to external thread	Gage
214.	Length of wrench flats from face	Gage
215.	Total length	Gage
216.	Radius of chamfer missing or incorrect	Visual
217.	Surface finish improper, except as otherwise classified	Visual

4.3.2.2 Retainer, Primer, Drawing B11731620, covering a detail of Drawing C11731585.

CRITICAL

1.	Metal defective	Visual

MAJOR

101.	True position of primer pocket with outside diameter	Gage
102.	True position of flash hole with outside diameter	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
103. Diameter, outside, max.	Gage
104. Diameter of primer pocket, max.	Gage
105. Depth of primer pocket	Gage
106. Total length	Gage
107. Protective finish inadequate or defective	Visual
108. Surface finish improper	Visual
109. Burr or sharp corner	Visual
110. Radius or chamfer missing	Visual
111. Foreign matter	Visual
MINOR	
201. Diameter of flash hole	Gage
202. Diameter, outside, min.	Gage
203. Diameter of primer pocket, min.	Gage
204. Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Primer, Percussion, M42, Drawing B8799925, covering a detail of Drawing C11731585.

CRITICAL	
1. Improper assembly	Visual
2. Any part missing	Visual
3. Primer pellet broken, cracked or dislodged	Visual
4. Metal defective	Visual
5. Primer foil discolored	Visual
6. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

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4.3.2.4 Retainer and Primer Assembly, Drawing C11731585, covering a detail of the applicable Delay Element Assembly Drawing.

Defect	Method of Inspection
CRITICAL	
1. Primer not properly seated in primer pocket	X-Ray 1/ and Visual
2. Improper assembly	X-Ray 1/ and Visual
3. Any part missing	X-Ray 1/ and Visual
4. Primer pellet broken, cracked or dislodged	Visual
5. Mixed primer	Visual
6. Metal defective	Visual
MAJOR	
101. Sealant missing	Visual
102. Primer crimp missing or incorrect	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Total length	Gage

1/ The retainer and primer assembly shall be X-Rayed with the assembly in a position perpendicular to the rays of the X-Ray machine. The X-Rays properly identified shall be forwarded to the procuring activity.

4.3.2.5 Assembly - Applicable Delay Element Assembly Drawing.

Defect	Method of Inspection
CRITICAL	
1. Improper assembly	X-Ray 1/ and Visual
2. Any part missing	X-Ray 1/ and Visual
3. Mixed ammunition	Visual
4. Flash hole missing or obstructed	Visual
5. Marking missing, incorrect or illegible	Visual
6. Metal defective	Visual
MAJOR	
101. Output charge not visible through flash hole	Visual

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
102.	Primer retainer not flush or below surface "S"	Visual
103.	Junction of primer retainer and delay body not properly sealed or crimped	Visual
104.	Protective finish inadequate or defective	Visual

MINOR

201.	Total length, max.	Gage
202.	Varnish missing	Visual

1/ The delay element shall be X-Rayed with the long axis of the delay element perpendicular to the rays of the X-Ray machine. The X-Ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.6 Container, Ammunition, Dwg. C9266885.

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Contents loose	Manual/Visual
202.	Quantity of contents incorrect	Visual
203.	Separator or filler missing	Visual
204.	Sealing inadequate, incomplete or incorrect	Visual
205.	Container cut or damaged	Visual
206.	Marking missing, incorrect or illegible	Visual

4.3.2.7 Box, Packing, Dwg. C9266886.

CRITICAL

None defined

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual/Visual
202. Hardware missing loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Contents incorrect	Visual
207. Marking missing, incorrect or illegible	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in MIL-STD-105, MIL-C-60539 and MIL-STD-171.

4.3.4 Primer sensitivity.

4.3.4.1 All-fire.-Fifty (50) primer retainer assemblies, shall be tested at the 13.0 inch height in accordance with 4.4.3. Any primer failing to comply with 3.4 shall be cause for rejection of the lot.

4.3.4.2 No-fire.-Fifty (50) primer retainer assemblies, shall be tested at the 2.0 inch height in accordance with 4.4.3. Any primer failing to comply with 3.4 shall be cause for rejection of the lot.

4.3.5 Vibration.-Twelve of the 30 assembled X-rayed delay elements of the initial production sample shall be subjected to vibration in accordance with 4.4.2. After vibration, the 12 vibrated delay elements shall be x-rayed as directed in 1/ 4.3.2.5. Failure of any delay element to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the lots of pellets, delay mix, primers or the method of charging, 12 additional delay elements shall be selected and tested as specified herein.

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4.3.6 Functional.-The following test shall be performed by the Government:

4.3.6.1 Ballistics.-Thirty test delay elements shall be grouped and temperature conditioned in accordance with 4.4.4.2 and Table IV. Failure of any delay element to comply with the requirements of 3.5.2 shall be cause for rejection of the lot.

Table IV

Temperature	Sample Size
-65 to -70F	10
65 to 75F	10
200 to 205F	10

4.3.7 Inspection equipment.-Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-A-2550.

4.4. Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539 and MIL-STD-171.

4.4.2 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Primer sensitivity.-The primer sensitivity test shall be made with Primer-Retainer assemblies. The test shall be made utilizing a 15/16 inch diameter steel ball weighing 1.94 plus/minus 0.02 ounces at each of the two heights. The test equipment shall have a firing pin protrusion of 0.045 plus/minus .002 inch. This shall be measured by seating the firing pin fully against the shoulder stop in the firing pin housing and measuring the resulting protrusion of the radius of the firing pin from the face of the test equipment firing pin housing. The contour of the firing pin striker, used in the test equipment to transmit energy to the primer, shall have a diameter of 0.080 inch and a spherical radius of 0.040 inch. The headspace between the primer retainer assembly and test equipment firing pin housing shall be zero. All height measurements shall be made between the head of the firing pin and the bottom of the suspended steel ball. The primer sensitivity test equipment shall be checked to insure a center impaction contact between the dropped steel ball and head of firing pin.

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4.4.4 Functional.

4.4.4.1 Ballistics.-Delay elements of the sample shall be grouped and temperature conditioned in accordance with 4.4.4.2 and Table IV and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.4.2.

4.4.2 Temperature conditioning.-An inert delay element containing a temperature sensitive device shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one (1) hour longer than the time necessary to bring the device contained in the inert delay element to the required temperature. The vent holes in the delay elements shall remain sealed against the environmental atmosphere of the conditioning chamber during the conditioning period. Upon completion of the temperature conditioning, the vent plugs shall be removed.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Dwgs. C9266885 and C9266886 except that the marking shall refer to the applicable Delay Element Assembly Specified in the contract or order (see 6.1 and 6.3).

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Delay time and the applicable Delay Element Assembly drawing number (see 2.1 and 6.3).

6.2 Delay time is defined as the time lapse between the impact of the firing pin on the primer and evidence of gas flow from the vent hole of the delay element.

6.3 Delay Element assemblies covered by this specification are identified as follows:

a. Delay Element Assembly 0.10 Second Delay Dwg. F11731469.

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6.1 (Cont'd)

b. Delay Element Assembly 0.30 Second Delay Dwg. F11738753.

c. Delay Element Assembly 0.60 Second Delay Dwg. F11731472.

d. Delay Element Assembly 0.85 Second Delay Dwg. F11731584.

e. Delay Element Assembly 1.0 Second Delay Dwg. F11731492.

6.4 Loaded Assembly Hazard Information.

Explosive Hazard Classification - Class 3
Storage Compatibility - Group B, E or N
Department of Transportation (DOT) - Hazard Class C
DOT Marking (Explosive Power Devices) - Class C (Keep Fire Away)

6.5 Hazard notice.-The Delay Element Assemblies described herein and certain of its components are flammable and explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing Activity:

Army - MU(FA)

Project No. 1377-A525

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DOCUMENT IDENTIFIER AND TITLE
MIL-D-50727B(MU) Delay Element Assemblies

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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